

Artificial Intelligence — 10-Page Master Quick Revision

High-Yield Formula Sheet, Complexity Matrix & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Artificial Intelligence (CS24307)

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⚡ Master Formula, Algorithm & Search Complexity Cheat Sheet

Algorithm / Concept	Core Mathematical Formulation / Rule	Key Exam Takeaway
Iterative Deepening Search (IDS)	Time: $O(b^d)$, Space: $O(bd)$	Linear space of DFS + Complete & Optimal like BFS.
A^* Evaluation Function	$f(n) = g(n) + h(n), \quad 0 \leq h(n) \leq h^*(n)$	Admissible $\implies$ Optimal Tree Search; Consistent $\implies$ Optimal Graph Search.
Alpha-Beta Pruning Rule	Prune if $\alpha \geq \beta$	Reduces search time from $O(b^m)$ to $O(b^{m/2})$ with perfect move ordering.
Bayesian Network Joint Formula	$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i))$	Drastically reduces probability parameters from $O(2^n)$ to $O(n \cdot 2^k)$.
Shannon Entropy	$H(S) = - \sum_{i=1}^c p_i \log_2(p_i)$	Measures data impurity; maximized at uniform probability distributions.
Information Gain	$\text{Gain}(S, A) = H(S) - \sum_v \frac{ S_v }{ S } H(S_v)$	Splitting criterion used in ID3 Decision Tree algorithm.
Perceptron Weight Update	$\mathbf{w} \leftarrow \mathbf{w} + \eta(y_{\text{true}} - y_{\text{pred}})\mathbf{x}$	Guaranteed to converge in finite steps for linearly separable datasets.

Top 10 High-Yield BIT Mesra Exam Questions & Solutions

Q1. State the 4 components of PEAS for an Automated Taxi Driver. (6 Marks)

- **P (Performance)**: Safety, speed, legal adherence, passenger comfort, profits.
- **E (Environment)**: Roads, traffic, pedestrians, weather, signals.
- **A (Actuators)**: Steering wheel, accelerator, brake, signal indicators, horn.
- **S (Sensors)**: Cameras, LiDAR, RADAR, GPS, speedometer, odometer, sonar.

Q2. Prove that if $h(n)$ is consistent, A^* using graph search never re-opens any node. (8 Marks)

By consistency, $h(n) \leq c(n, a, n') + h(n') \implies g(n) + h(n) \leq g(n) + c(n, a, n') + h(n') = g(n') + h(n') \implies f(n) \leq f(n')$.

Thus, $f(n)$ values along any path are monotonically non-decreasing. When A^* selects node n from the frontier for expansion, the optimal path cost to n has already been found. Hence, no shorter path to n can be discovered later, making re-opening closed nodes unnecessary. ■